

Daily Review

Booklet p.3 — The Solar System and Beyond

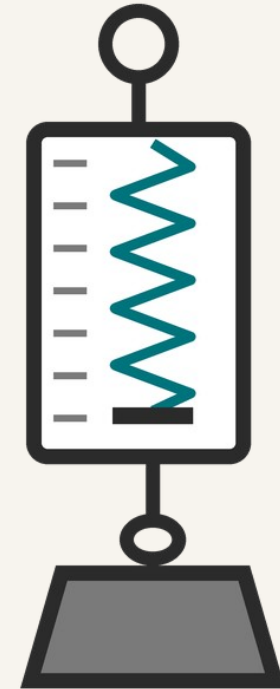
- 1 What is a moon?
A natural satellite of a planet
- 2 Name the galaxy we live in.
The Milky Way
- 3 What is happening at the core of a star?
Nuclear fusion
- 4 Put in order of size: planet, asteroid, moon.
Asteroid, moon, planet
- 5 What is an exoplanet?
A planet orbiting another star

SPACE

Mass and Weight

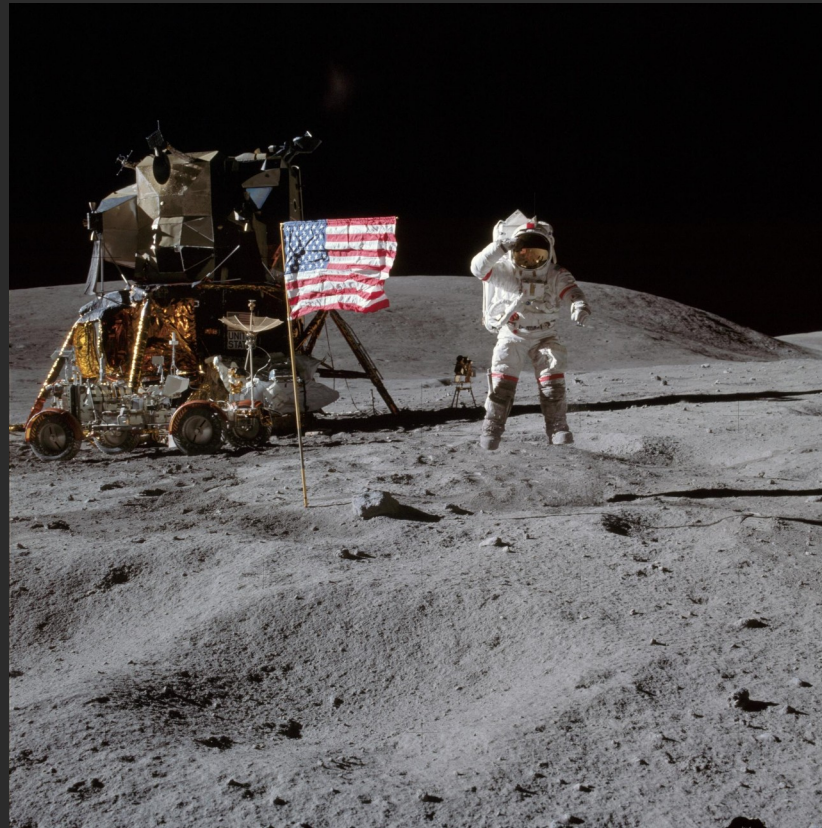
Your mass goes wherever you go. Your weight does not.

National 5 Physics (S3)



THE BIG QUESTION

An astronaut jumps much higher on the Moon. What has changed — and what has not?



John Young, Apollo 16, 1972 — NASA

PART 1

Mass and weight

Two words everyone mixes up — and why they are different.

By the end of this part...

- 1 I can describe the difference between mass and weight

Show me — Describe each word

SHOW ME BOARDS

1 mass

The amount of matter in an object

2 weight

The force due to gravity on an object

3 gravitational field strength

The weight per kilogram of mass

4 kilogram

The unit of mass (kg)

5 newton

The unit of force and weight (N)

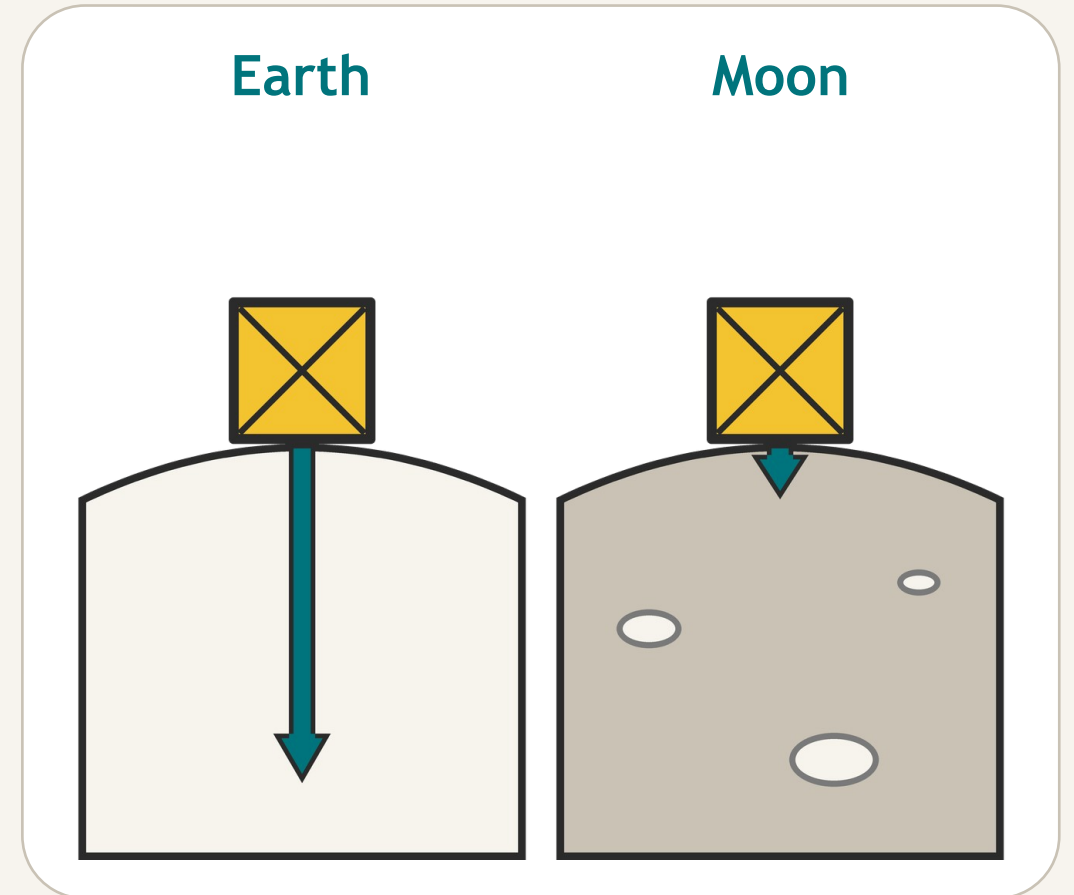


The Moon — NASA / photographed from lunar orbit on the Apollo 15 mission

Same crate, two places

Booklet p.5

- Mass is the amount of matter in an object.
- It is measured in kilograms, and stays the same everywhere.
- Weight is the force due to gravity on the object.
- It is measured in newtons, and is different on every planet.



Measuring mass in orbit

On the International Space Station



- Scales read zero in orbit: both fall together.
- This device's springs push with a known force.
- The acceleration gives the astronaut's mass.
- Mass is the same in orbit as on Earth.

Show me — mass or weight?

SHOW ME BOARDS

- 1 Which one is measured in newtons?
- 2 Which one stays the same on the Moon?
- 3 An astronaut flies to Mars. What happens to their weight?
- 4 What is the unit of mass?
- 5 Is 'my weight is 60 kg' correct? Why?

Weight

Mass

It decreases

kilogram (kg)

No: 60 kg is a mass

Show me — mass or weight?

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Which one is a force?
- 2 Which one changes from planet to planet?
- 3 A rock is brought back from the Moon. What happens to its mass?
- 4 What is the unit of weight?
- 5 Which one could you measure with a spring balance?

Weight

Weight

It stays the same

newton (N)

Weight

Agree, disagree or it depends?

SHOW ME BOARDS

Boards up: AGREE, DISAGREE or IT DEPENDS — and why

- 1 An astronaut has less mass on the Moon.
Disagree: mass never changes
- 2 Weight is measured in kilograms.
Disagree: weight is a force, in N
- 3 Astronauts in orbit have no weight.
Disagree: gravity acts
- 4 A bag weighs more on Earth than on Mars.
Agree: g is bigger on Earth
- 5 A heavier object has more mass.
It depends: only in the same place

Correct your booklet

Booklet p.5, Task 1

CROSS OUT AND FIX

Mass: cross out “and object is made off”.
Write: “an object is made of”.

Task 1 – complete the table

Booklet p.5

Fill in the blanks for mass and for weight.

Booklet p.5, Task 1

Task 1 – Answers

ANSWERS

Booklet p.5

Term	Accepted wording
Mass	The amount of matter an object is made of.
	Stays the same on every planet.
Weight	The force due to gravity on an object.
	Is different on every planet.

The relationship

Booklet p.5 – Task 2

$$W = m g$$

W

weight, in
newtons (N)

m

mass, in kilograms
(kg)

g

gravitational field
strength, in N kg^{-1}

Task 2 – label the equation

Booklet p.5

Label each letter in $W = mg$
with its quantity and its unit.

Booklet p.5, Task 2

Task 2 – Answers

ANSWERS

Booklet p.5

Letter	Quantity	Unit
W	weight	newton (N)
m	mass	kilogram (kg)
g	gravitational field strength	N kg^{-1}

Daily Review

Booklet p.5 — Part 1

- 1 What is mass?
- 2 What is weight?
- 3 What is the unit of weight?
- 4 Which one stays the same on every planet?
- 5 Write the relationship for weight.

The amount of matter in
an object

The force due to gravity
on an object

newton (N)

Mass

$$W = mg$$

PART 2

Weight, mass and field strength

One relationship for weight — anywhere in the universe.

By the end of this part...

- 1 I can solve problems with weight, mass and field strength
- 2 I can use the data sheet for different places in the universe
- 3 I can read field strength from a graph

g is different in different places

Data sheet, booklet p.2

Place	$g \text{ (N kg}^{-1}\text{)}$	Place	$g \text{ (N kg}^{-1}\text{)}$
Earth	9.8	Venus	8.9
Moon	1.6	Uranus	8.7
Mars	3.7		

Look the value up every time — never assume 9.8.

Correct your booklet

Booklet p.2, data sheet

CROSS OUT AND FIX

Mercury: cross out 4.0 N kg^{-1} .

Write 3.7 N kg^{-1} — the value on every SQA data sheet.

Worked example 1

Finding a weight

A lunar rover has a mass of 900 kg.
Calculate its weight on the Moon.

$$m = 900 \text{ kg}$$

$$g = 1.6 \text{ N kg}^{-1}$$

$$W = ?$$

$$W = m g$$

$$W = 900 \times 1.6$$

$$\underline{W = 1440 \text{ N}}$$

Your turn 1 — a rover on Mars

SHOW ME BOARDS

A Mars rover has a mass of 1200 kg.
Calculate its weight on Mars.

$$m = 1200 \text{ kg}$$

$$g = \dots\dots \text{ N kg}^{-1}$$

$$W = ?$$

$$W = m g$$

$$W = 1200 \times \dots\dots$$

$$\underline{W = \dots\dots \text{ N}}$$

$$g = 3.7 \text{ N kg}^{-1}$$

$$W = 4440 \text{ N}$$

Worked example 2

Rearranging for mass

A toolbox has a weight of 178 N on the surface of Venus.
Calculate its mass.

$$W = 178 \text{ N}$$

$$g = 8.9 \text{ N kg}^{-1}$$

$$m = ?$$

$$W = m g$$

$$m = W/g$$

$$m = 178/8.9$$

$$\underline{m = 20 \text{ kg}}$$

Your turn 2 – a crate on Uranus

SHOW ME BOARDS

A crate has a weight of 348 N on Uranus.
Calculate its mass.

$$W = 348 \text{ N}$$

$$g = \dots\dots \text{ N kg}^{-1}$$

$$m = ?$$

$$g = 8.7 \text{ N kg}^{-1}$$
$$m = 40 \text{ kg}$$

$$W = m g$$

$$m = W/g$$

$$m = 348/\dots\dots$$

$$\underline{m = \dots\dots \text{ kg}}$$

Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

A robot of mass 25 kg is taken to Mars.
Its weight on Mars is

- A 3.7 N
- B 6.8 N
- C 28.7 N
- D 92.5 N
- E 245 N.

Answer D

Show me — $W = mg$

SHOW ME BOARDS

EXTRA PRACTICE

1

A 65 kg astronaut is on the Moon.
Calculate their weight.

104 N

2

A 3.0 kg bag is on Uranus. Calculate its
weight.

26.1 N

3

An object weighs 98 N on Earth.
Calculate its mass.

10 kg

4

A 200 kg lander weighs 1780 N. Which
planet is it on?

Venus ($g = 8.9 \text{ N kg}^{-1}$)

5

Which weighs more: 10 kg on Earth or
40 kg on the Moon?

10 kg on Earth
98 N vs 64 N

Watch out

THE MISTAKE

- “My weight is 60 kg”
- “Gravity on Mars is 3.7”
- “Going to the Moon changes my mass”

THE CORRECT ANSWER

- My mass is 60 kg; my weight is 588 N
- Gravitational field strength is 3.7 N kg^{-1}
- Mass stays the same; only weight changes

Demo – the newton balance

DEMO

A 1.0 kg mass hangs on a newton balance.
Predict the reading – then read it.

Does the balance measure mass or weight?
Predict its reading on the Moon.

Task 3 – try the examples

Booklet p.5

Use the data sheet on p.2.
Set each one out like the worked examples.

Booklet p.5, Task 3

Task 3 – Answers

ANSWERS

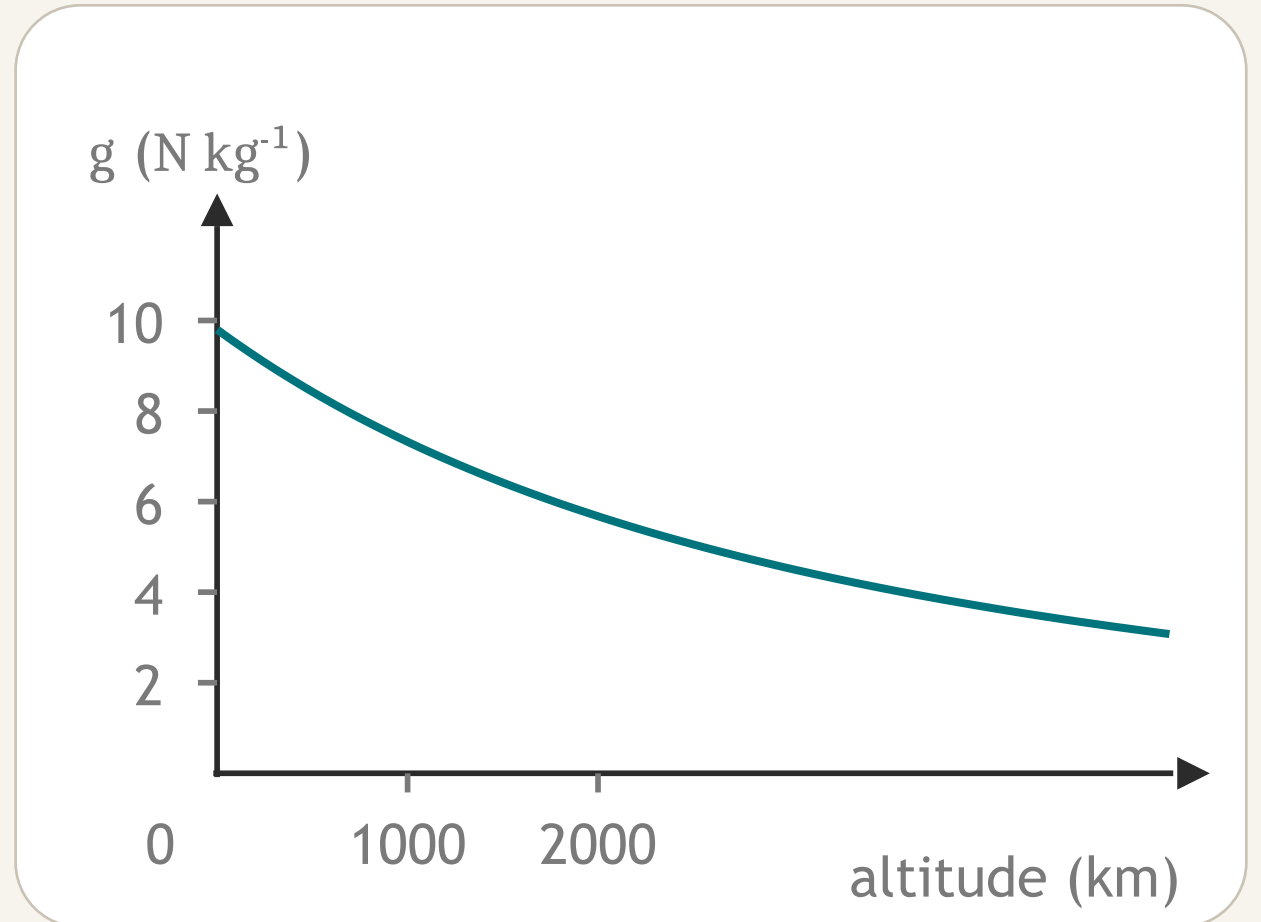
Booklet p.5

Example	Working	Answer
Jimmy, on Earth	$W = mg = 75 \times 9.8$	735 N
Sally, on Mars	$W = mg = 60 \times 3.7$	222 N
The rock	$m = W \div g = 196 \div 9.8$	20 kg

Higher up, a weaker field

Above the surface of the Earth

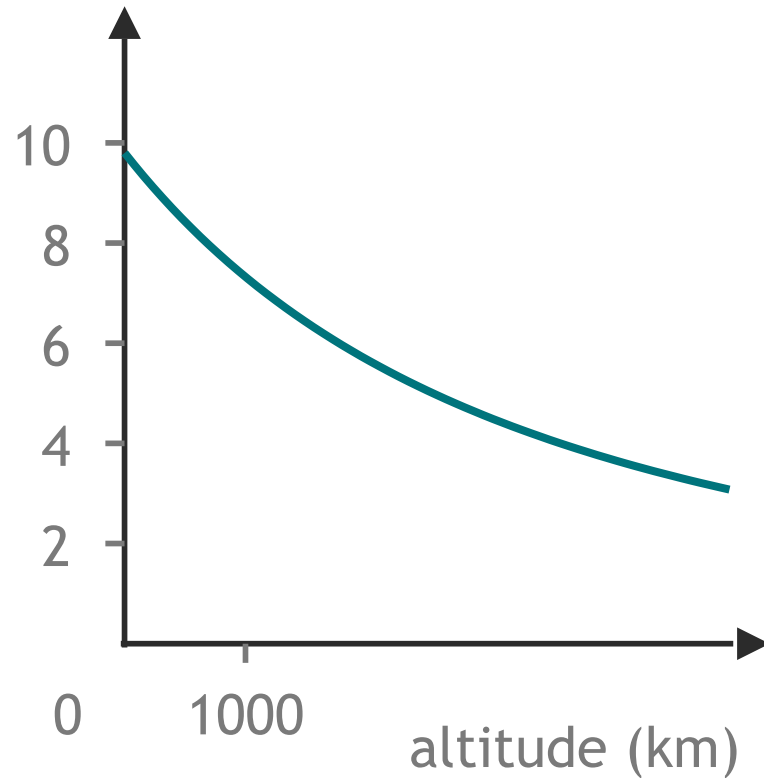
- The further from the planet, the smaller g is.
- Mass does not change with height.
- So weight gets smaller as an object climbs.



Show me — use the graph

SHOW ME BOARDS

$g \text{ (N kg}^{-1}\text{)}$



1

What is g at the surface?

9.8 N kg^{-1}

2

What happens to g higher up?

It decreases

3

A satellite climbs to 2000 km. Its mass?

Stays the same

4

...and its weight?

It decreases

Checkpoint

- 1 What is the difference between mass and weight?
Mass is matter (kg); weight is a force (N)
- 2 A 30 kg dog is on Mars. Calculate its weight.
111 N
- 3 Why does an astronaut's weight change on the Moon, but not their mass?
g is smaller; the matter is the same

Exam practice

Past paper

- National 5 Physics 2023, Section 2, question 3(b). 3 marks.
- The weight of a small satellite at its orbital altitude.
- Then booklet p.6-7, Task 4.

LOOKING AHEAD

Space Exploration and the ISS

Living in orbit is dangerous. How do astronauts survive it?

Plenary

- 1 The big question: why can an astronaut jump higher on the Moon?
- 2 What is mass, and what is its unit?
- 3 What is weight, and what is its unit?
- 4 A 40 kg crate is on Earth. Calculate its weight.
- 5 An object weighs 16 N on the Moon. Calculate its mass.

Same mass, smaller g :
less weight

Amount of matter; kg

Force due to gravity; N

392 N

10 kg

Plenary

EXTRA PRACTICE

- 1 Which stays the same everywhere — mass or weight?
- 2 State the relationship between weight, mass and g .
- 3 A 5.0 kg bag is on Venus. Calculate its weight.
- 4 A rover weighs 740 N on Mars. Calculate its mass.
- 5 A satellite climbs higher. What happens to its weight?

Mass

$$W = mg$$

44.5 N

200 kg

It decreases

Extension

- 1 A 70 kg astronaut weighs 112 N. Where are they standing?
- 2 Is it easier to lift a 100 kg mass on the Moon or on Earth? Explain.
- 3 A rocket burns fuel as it climbs. Give two reasons its weight falls.

Extension

EXTRA PRACTICE

- 1 An object weighs 44.5 N on Venus. What is its mass on the Moon?
- 2 A 60 kg astronaut flies to the Moon. By how much does their weight fall?
- 3 Why is it wrong to say an astronaut in orbit has no weight?

Course specification

N5 Physics course specification v4.0, p.8 – Space exploration

- Use of an appropriate relationship to solve problems involving weight, mass and gravitational field strength, in different locations in the universe.
- $W = mg$